

CV2090
Probability & Statistics for
Civil Engineers
(Jul – Nov 2026)

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L15
Aug 24, 2026 (3 pm)

Announcement

- 24th Aug (today):
 - Class problems to replace tutorials today
- 25th Aug: No class on Tuesday
- 31st Aug: Quiz-1 (8 AM)
 - Quiz-1 syllabus: up to random variables (including Normal and χ^2 , but excluding *functions of rv*)
- Check course page on GitHub regularly
<https://github.com/prakashvs613/2026-CV2090-IITM/>

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Messages from the Last Class

- **Message 1:** Normal distribution

→ Gaussian/Normal/ Bell-shaped

→ PDF, $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}$

$N(\mu, \sigma^2)$

→ When σ there are additive effects.

→ CDF does not have a closed-form solution
→ use standard tables

→ Smooth, 2 points of inflection

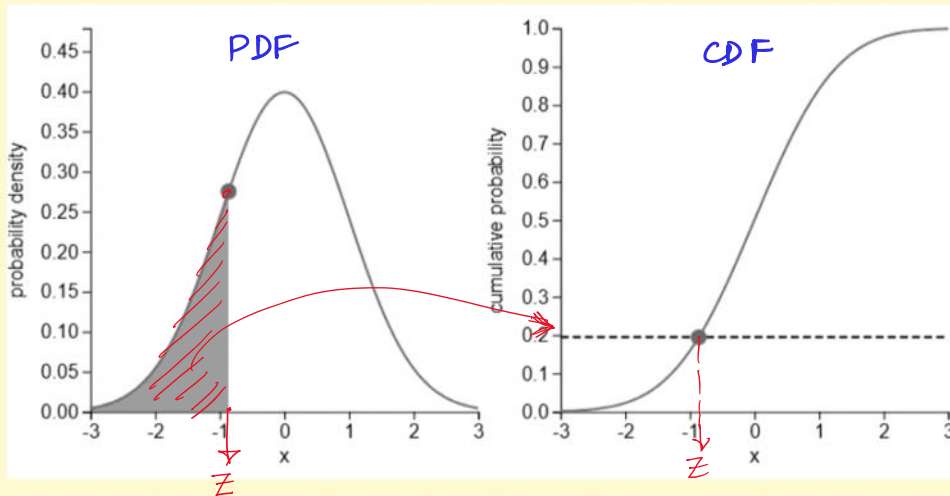
→ Completely defined by its mean & variance, μ, σ^2

→ 68-95-99.7 rule

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Messages from the Last Class

• Message 1: Normal distribution



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Class problems

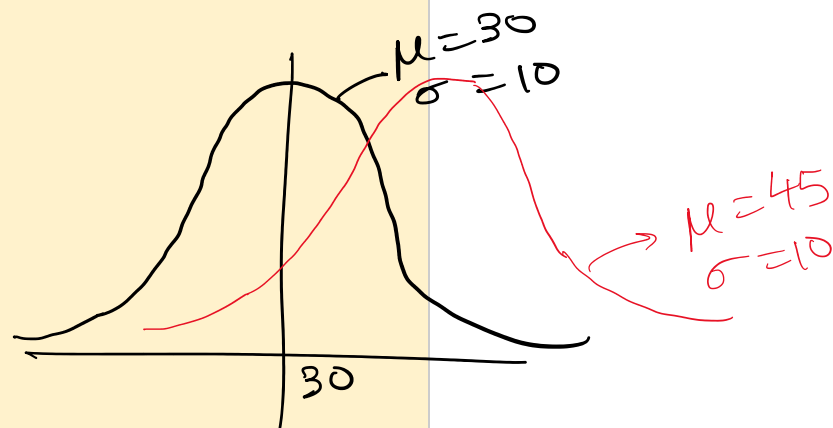
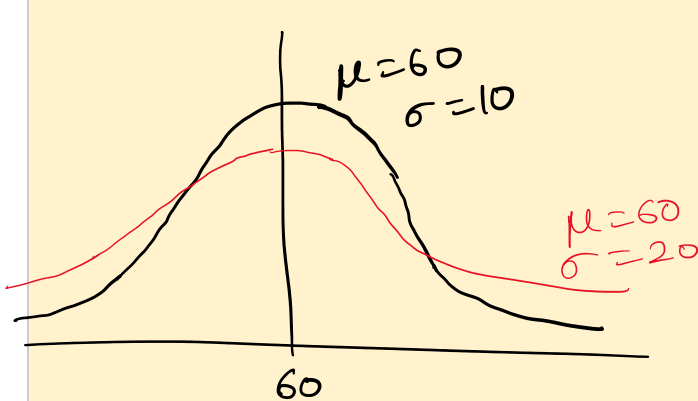
Draw Gaussian PDFs for

1. Varying σ

- $\mathcal{N}(\mu = 60, \sigma = 10)$, $\mathcal{N}(\mu = 60, \sigma = 20)$, $\mathcal{N}(\mu = 60, \sigma = 30)$

2. Varying μ

- $\mathcal{N}(\mu = 30, \sigma = 10)$, $\mathcal{N}(\mu = 45, \sigma = 10)$, $\mathcal{N}(\mu = 60, \sigma = 10)$



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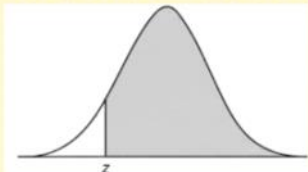
Normal distribution

- Three types of areas under the curve are useful



$Pr(Z \leq z)$

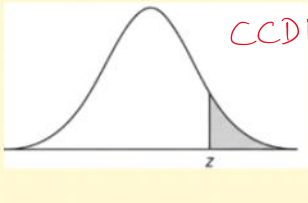
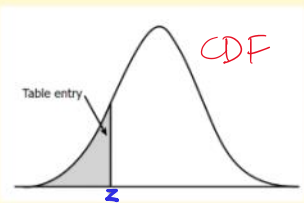
- Same as CDF
- Read from the table
- if $z > 0$, $Pr(Z \leq z) > 0.5$



$Pr(Z > z)$
 $= 1 - Pr(Z \leq z)$
 $= 1 - \Phi(z)$
 • if $z > 0$
 $\Rightarrow Pr(Z > z) < 0.5$

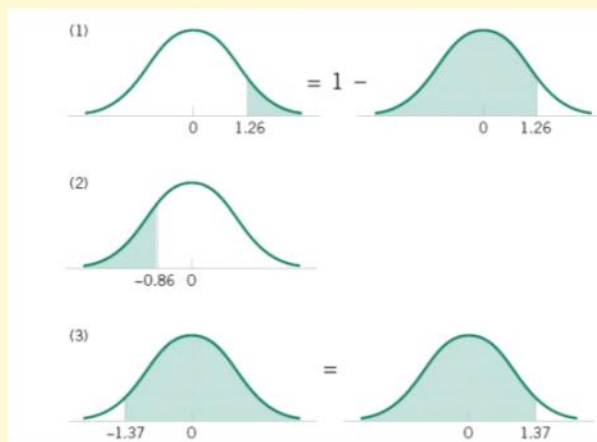


$Pr(-z \leq Z \leq z)$
 $= Pr(Z \leq z) - Pr(Z \leq -z)$
 $= 1 - 2 \cdot Pr(Z \leq -z)$
 $= 2 \cdot Pr(Z \leq z) - 1$



68 - 95 - 99.7
 ± 1 ± 2 ± 3
 interval

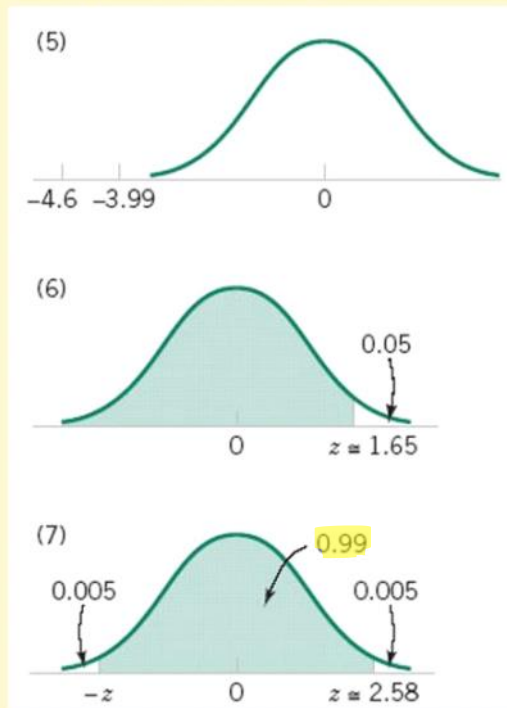
Examples (using basic prob. & symmetry)



$$Pr(-1.25 < Z < 0.37) = Pr(Z < 0.37) - Pr(Z \leq -1.25)$$

$$= 0.6443 - 0.1058$$

Examples (using basic prob. & symmetry)



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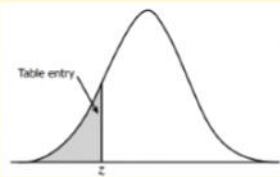
Class problems

- **Example 1:** Given a factory machine filling cereal boxes with a mean weight $\mu = 500$ g and $\sigma = 10$ g, find the probability that a randomly chosen box weighs less than 485 g
- **Example 2:** Find the threshold weight that marks the top 10% of heaviest boxes.

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Standard Normal Probabilities

Z-table



$$\Phi(z) = \Pr(Z \leq z)$$

Table entry for z is the area under the standard normal curve to the left of z .

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

Class problems

- **Example 3:** A concrete batch has a mean strength of 10 MPa; variance 4 MPa²
- What is the probability that a test result exceeds 13 MPa?

Standard Normal RV

In last few examples,

$$\begin{array}{ccc} \mathcal{N}(\mu, \sigma^2) & \xrightarrow{\quad} & Z(0, 1) \\ \uparrow & & \\ X & & \end{array}$$
$$Z = \frac{X - \mu}{\sigma}$$

PDF

standard Normal
Random Variable

$$\phi(z) = f_Z(z) = \frac{1}{\sqrt{2\pi}} \cdot e^{-z^2/2}$$

$$\Phi(z) = \Pr(Z \leq z)$$

→ CDF

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Standardizing a Normal RV

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Class problems

- **Example 4:** A delivery company operates an elevator designed to carry a maximum weight limit of **500 kg**. A group of **4 random men** board the elevator. Their weights are normally distributed with a mean (μ) of **75 kg** and a standard deviation (σ) of **10 kg**.
- Calculate the probability that the total weight of these 4 men *exceeds* the 500 kg safety limit.

If X_1, X_2, X_3, X_4 are Independent NRV

$$\text{Sum, } S = X_1 + X_2 + X_3 + X_4$$

$$\mu_S = 4\mu_X$$
$$\Rightarrow \mu_S = 300 \text{ kg}$$

$$\sigma_S^2 = \sigma_X^2 + \sigma_X^2 + \sigma_X^2 + \sigma_X^2$$
$$= 4\sigma_X^2$$

$$\Rightarrow \sigma_S = 2\sigma_X = 20 \text{ kg}$$

$$\Pr(S > 500 \text{ kg}) = \Pr\left(Z > \frac{500 - 300}{20}\right) = \Pr(Z > 10)$$
$$\approx 0$$

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Class problems

if X_1, X_2, X_3, X_4 are normal and independent

What about $S' = 4X_1$? Determine the mean and stdev of S' ?

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Class problems

- **Example 5:** Professor A's midterm has a mean of $\mu = 50$ and $\sigma = 20$. Professor B's midterm has a mean of $\mu = 70$ and $\sigma = 5$. A student scores a **65** on Professor A's test and a **72** on Professor B's test. Both classes grade on a strict curve where top 7.5% get an S.
- Find out in *which* class the student actually earned a better relative standing (higher percentile).

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Lognormal Distribution

Ang and Tang (Section 3.2.2)